



BASE24-pos[®]

Multiple Currency Support Manual

© 2003 by ACI Worldwide Inc. All rights reserved.

All information contained in this document is confidential and proprietary to ACI Worldwide Inc. This material is a trade secret and its confidentiality is strictly maintained. Use of any copyright notice does not imply unrestricted or public access to these materials. No part of this document may be photocopied, electronically transferred, modified, or reproduced in any manner without the prior written consent of ACI Worldwide Inc.

ACI, BASE24, BASE24 Remote Banking, BASE24-atm, BASE24-atm self service banking, BASE24-billpay, BASE24-card, BASE24-from host maintenance, BASE24-InfoBase, BASE24-mail, BASE24-pos, BASE24-telebanking, BASE24-teller, InfoBase, NET24-XPNET, and Perfmon are trademarks or registered trademarks of ACI Worldwide Inc., Transaction Systems Architects, Inc., or their subsidiaries.

Other companies' trademarks, service marks, or registered trademarks and service marks are trademarks, service marks, or registered trademarks and service marks of their respective companies.

Contents

Preface	v
 Section 1	
Introduction	1-1
Multiple Currency Support	1-2
Supported Transactions, Message Types, and Authorization Levels	1-4
Files Maintenance	1-5
Exchange Rate File (ERF)	1-5
Other Files Maintenance Information	1-6
Tokens	1-8
LCONF Assigns and Params	1-9
Interfaces with Host Systems	1-10
ISO Host Interface Process	1-10
PTLF Extract	1-10
PBF Refresh	1-11
CAF Refresh	1-11
POS Device Handler Processing	1-12
 Section 2	
Currency Conversion Processing	2-1
Currency Types	2-2
Currency Types Held in BASE24-pos Files	2-4
Currency Conversion	2-5
Currency Conversion Utilities	2-6
Currency Conversion Flows	2-7
BASE24-pos Authorizes the Transaction	2-8
A Host Authorizes the Transaction	2-10
An Interchange Authorizes the Transaction	2-12
An Interchange Is the Acquirer	2-15

Section 3

Authorization Processing. 3-1

 Overview 3-2

 Currency Types 3-4

 Initialization and Warmboot Processing 3-6

 Preauthorization Processing 3-7

Index. Index-1

Preface

The *BASE24-pos Multiple Currency Support Manual* provides a discussion of the aspects of the Multiple Currency add-on product that pertain to the BASE24-pos product.

Audience

This manual is intended for personnel who must understand the implementation of the BASE24-pos Multiple Currency add-on product.

Prerequisites

This manual assumes that you are very familiar with normal BASE24-pos processing.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The *BASE24 Base Files Maintenance Manual* provides a comprehensive explanation of the BASE24 screens that access the base and interchange interface files.
- The *BASE24 Logical Network Configuration File Manual* describes the BASE24 assigns and params.

- The ***BASE24 Tokens Manual*** contains specifications for tokens and token processing.
- The ***BASE24-pos CRT Authorization and CRT Administration Manual*** describes the Multiple Referrals screen that is used with the CRT Authorization add-on product.
- The ***BASE24-pos Files Maintenance Manual*** provides descriptions of the BASE24 screens that access BASE24-pos files and functions.
- The ***BASE24-pos Transaction Processing Manual*** describes how BASE24 processes standard BASE24-pos messages.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in This Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” provides an overview of the BASE24-pos Multiple Currency add-on product.

Section 2, “Currency Conversion Processing,” provides a discussion of the various currency types used in processing Multiple Currency transactions and how the conversions are processed from one type to another.

Section 3, “Authorization Processing,” describes how the Multiple Currency add-on product affects the BASE24-pos Authorization module.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, PS-AE300-10 for the *BASE24-pos Multiple Currency Support Manual*) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, 11/2003 for November, 2003) indicates the issue of the manual. The software release information (for example, R6.0v4 for release 6.0, version 4) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

ACI Worldwide Inc.

Section 1

Introduction

The BASE24-pos Multiple Currency add-on product provides multiple currency support for BASE24-pos by managing the processing of a mixture of currencies involved in an individual transaction.

The Multiple Currency product allows institutions to offer accounts in different currencies, with the ability to define each account with its own currency. It also allows cardholders to perform transactions in different currencies at different POS devices.

In addition, BASE24-pos can acquire transactions in different currencies from a single POS device. The Standard POS Device Handler (SPDH) and APACS 30/40 Device Handler modules provide support for terminals that allow transactions to be performed in different currencies.

Multiple Currency Support

Within a BASE24-pos Multiple Currency system, any number of currencies can be supported, although a maximum of ten different currency types, other than the Transaction currency, can be used in the processing of a single transaction. Currency conversion is performed as necessary providing the following capabilities:

- Transactions can be designated in any currency that is supported in the logical network.
- POS devices can be configured to operate in a currency that is different from the default currency of the owning institution, and the same or different from that of another POS device belonging to the same institution.
- Certain POS devices support transactions specified in different currencies, with the currency of the transaction selected at the point of sale.
- Customer accounts can be designated in any supported currency.
- Where multiple accounts are associated with a single card, these accounts can be in different currencies.
- Transactions performed in one currency can be processed against an account held in a different currency.
- Limit and usage checking can be performed regardless of the currency of the transaction.
- All currencies used during processing of a single transaction are stored in a BASE24 token and logged to the POS Transaction Log File (PTLF), including the equivalent amounts in each currency used. This information can therefore be made available to external interfaces and extracted for host processing if required.

BASE24-pos supports transaction processing in multiple currencies by defining currency types and allowing these to be used in data files and financial messages within BASE24. Some of these currency types are briefly defined in the following list. Section 2 describes these currency types in more detail.

Transaction Currency — The currency in which the transaction is delivered to BASE24 from a device, a host, or an interchange. This is the key currency for multiple currency processing.

Acquirer Institution Currency — The default currency for the institution that acquires the transaction.

Issuer Institution Currency — The default currency for the institution that is the issuer for the transaction.

Account Currency — The designated currency for the account(s) to which the transaction applies.

Base Currency — The currency through which all currency conversion takes place. All exchange rates are defined in the Base currency.

Original Currency — The currency used at the device at which the transaction originates. If the transaction is acquired by BASE24, Original currency is the same as the Transaction currency. However, if the transaction enters BASE24 through an interchange, the Original currency may be different than any other currency known to BASE24.

Terminal Currency — The currency for the device at which the transaction is performed.

Logical Network Currency — Logical Network currency is maintained in the Multiple Currency product primarily to provide compatibility with standard BASE24 processing for a number of subsidiary functions.

Any process performing multiple currency processing converts the Transaction currency into these various currencies using the Currency Conversion utilities. Currency conversion processing is discussed in section 2.

Supported Transactions, Message Types, and Authorization Levels

The BASE24-pos Multiple Currency add-on product supports the following transactions, message types, and authorization levels.

Transaction Types	All transaction types defined for the BASE24-pos product
Message Types	All message types defined for the BASE24-pos product, except for chargeback request and response
Authorization Levels and Methods	All authorization levels and methods defined for the BASE24-pos product, except for Parametric Authorization

For more information about transaction types, message types, and authorization levels, see the ***BASE24-pos Transaction Processing Manual***.

The BASE24-pos Multiple Currency add-on product supports the standard set of BASE24-pos response codes detailed in the ***BASE24-pos Transaction Processing Manual***. Any transaction rejected due to currency conversion errors is reported using one of the standard response codes (e.g., 101 — issue a call).

Files Maintenance

One BASE24 file is specific to the Multiple Currency product: The Exchange Rate File (ERF) contains one record for each currency, except the Base currency, that can be used by BASE24. In addition, some fields in other files may have additional meaning when used with the Multiple Currency product.

For information about the screens, see the *BASE24 Base Files Maintenance Manual*, *BASE24-pos Files Maintenance Manual*, and *BASE24-pos CRT Authorization and CRT Administration Manual*.

Exchange Rate File (ERF)

The Exchange Rate File (ERF) contains one record for each currency (except the Base currency) that is used by BASE24. The Base currency in multiple currency processing is the single currency against which exchange rates for all other currencies are expressed. All exchange rates held in the ERF are in respect of Base currency to a specific currency rate. That is, the value by which the amount must be multiplied or divided to obtain the equivalent amount in the specific currency.

Records are added to the ERF using the BASE24 files maintenance facility. No historical rates are stored in the ERF. However, a history of previous exchange rates can be derived by analyzing data in the Online Maintenance File (OMF).

There is one ERF per logical network and the file must contain at least one record if BASE24 is processing multiple currencies.

Other Files Maintenance Information

The following table provides information on additional screens in which some fields may have additional meaning when used with the Multiple Currency product:

Institution Definition File (IDF)	The CURRENCY CODE field on screen 3 (i.e., the Institution currency) can be modified to be different than any other Institution currency or different from the Logical Network currency. The restriction that this field must contain the same value as the LCONF CURRENCY-CODE param no longer applies. The value of the Currency Code does not have to be the same for all IDF records in the same Refresh group. For acquiring interchange interfaces (i.e., those that deliver transactions to BASE24), an IDF record must exist for each FIID specified in the Interchange Configuration File (ICF) or the Enhanced Interchange Configuration File (ICFE). This is required for BASE24 to derive the Acquiring Institution currency for a transaction.
Card Prefix File (CPF)	You can configure standard increment amounts for credit account withdrawals (cash advance) on screen 6 of the CPF. However, these will be successful only if the Transaction currency is the same as the Issuer Institution currency. Cash advance transactions performed in any other currency will be denied with a response code indicating an invalid cash amount. To avoid this, a standard increment value should not be configured in the CPF.
Positive Balance File (PBF)	The CURRENCY CODE field indicates the Account currency of the specified PBF record. If the currency code is updated, BASE24 performs no recalculation of balances or amounts.
POS Transaction Log File (PTLF)	The CURR CODE field on the PTLF Summary screen shows the alphabetic currency code, in the Transaction currency, for the corresponding AMOUNT field. In a Multiple Currency environment, several currency codes could be displayed on this screen.

BASE24-pos Terminal Data files (PTD)	The CURRENCY CODE field on screen 2 indicates the ISO numeric code identifying the primary currency used by the terminal. If no value is entered in this field, it defaults to the value of the first entry in the CURRENCY-CODE-TABLE in the COBNAMES File. The MC TOTALLING TYPE field on screen 2 indicates whether terminal totals in a multiple currency environment are stored in the PTD as hash totals or in the primary Terminal currency.
Multiple Referrals screen	The CURR CODE field indicates the Transaction currency in which the transaction was processed. This screen is used with the CRT Authorization add-on product only.
POS Referral File (PRF) screen	The CURRENCY CODE field identifies the type of Transaction currency being used in the transaction. This screen is used with the CRT Authorization add-on product only.

Tokens

Three tokens facilitate BASE24-pos support of multiple currencies:

- Multiple Currency token
- POS Balances token
- Original Currency 6.0 token

The following table provides a brief description of each token. For more information about BASE24 tokens, including descriptions of the tokens used by the BASE24-pos Multiple Currency Support product, see the ***BASE24 Tokens Manual***.

Token Name	Token ID	Description
Multiple Currency token	BD	Currency-related data that has been accumulated during the processing of a transaction. This token provides additional data relating to the transaction for extract and reporting purposes, as well as for sending online to a host.
POS Balances token	CB	Cardholder account balance data when there is a requirement for the account balance to be displayed or printed at the POS device in the Account currency.
Original Currency 6.0 token	BE	Data relating to a currency conversion that has been performed by an interchange before presenting the transaction to BASE24. That is, the currency in which the transaction is presented to BASE24 is different from the originating currency. BASE24-pos performs no additional processing on this data.

LCONF Assigns and Params

The BASE24 Logical Network Configuration File (LCONF) enables institutions to define files, processes, and spooler locations at the network, product, process, and device level. The LCONF also enables institutions to define processing parameters at the network, product, process, and device level. Institutions make these definitions by configuring assigns and params in the LCONF.

The Multiple Currency product uses the following assigns and params:

- The ERF assign defines the location of the Exchange Rate File (ERF). This assign is used by the ERF Extended Memory Table Build process.
- The ERFEMT assign defines the location of the Exchange Rate File (ERF) extended memory table to be used by this logical network. This assign is used by the Currency Conversion utilities and the Extended Memory Table Build process.
- The BASE-CURRENCY-CODE param specifies the ISO currency code of the Base currency to be used by this logical network.
- The CURRENCY-CODE param specifies the primary type of currency the logical network uses for activity limits established in the Card Prefix File (CPF).
- The ERFEMT-MAX-TBL-SIZE param specifies the maximum size in megabytes of the ERF extended memory table.
- The ERFEMT-WARM*Identifier* param specifies the symbolic name of the BASE24-pos Router/Authorization module to be warmbooted when the ERF extended memory table is warmbooted from the Device Control Terminal (DCT).

For details of the LCONF assigns and params, including those used by the BASE24-pos Multiple Currency product, see the ***BASE24 Logical Network Configuration Manual***.

Interfaces with Host Systems

Host systems can be impacted by implementation of the BASE24-pos Multiple Currency product in a number of ways. The precise impacts depend on the authorization methods used and the type of multiple currency functionality to be supported.

ISO Host Interface Process

Host systems must be aware that the Multiple Currency product is supported. The amount and currency fields in the BASE24 external message conform to the ISO 8583 (1987) standard, *Bank Card Originated Messages—Interchange Message Specifications—Content for Financial Transactions*. The Transaction currency defined by BASE24 is used for all standard amounts in requests, responses, reversals, and other transaction messages.

In a multiple currency system, the currency code in the Institution Definition File (IDF) is used to define the Issuer Institution currency, which may not be the currency used in the request message. Although only one currency is defined as the Transaction currency in a single transaction, issuer hosts may receive different authorization requests in different currencies.

An IDF record including a currency code must be configured for the FIID associated with each acquiring host. This is required to establish the Acquirer Institution currency. Although only one currency is defined as the Transaction currency in a single transaction, acquiring hosts may deliver different authorization requests to BASE24-pos in different currencies.

BASE24 tokens can be appended to the ISO messages. The Multiple Currency, POS Balances, and Original Currency 6.0 tokens can be used to convey the transaction amounts in all applicable currencies, to and from host systems, if required for host processing.

PTLF Extract

The Transaction currency defined by BASE24 is used for all standard amount and currency fields in the PTLF. The primary amounts are logged to the PTLF in the Transaction currency and included in the standard extract format. PTLF extracts are generated with amount fields in the Transaction currency.

The Multiple Currency, POS Balances, and Original Currency 6.0 tokens can also be extracted, according to configuration and availability. Therefore, these tokens can be used to convey the transaction amounts in all applicable currencies to host systems, if required for host processing.

PTLF Extract files may contain records in different currencies, as the BASE24-pos system can process transactions (and settlement data) that are acquired in multiple currencies.

During cutover from one business day to the next, the Settlement process writes terminal totals records to the PTLF. The settlement totals include a currency code indicator. PTLF extracts are generated with amount fields in the Transaction currency.

PBF Refresh

Cardholders can hold accounts in a choice of currency. The Positive Balance File (PBF) Refresh format for release 6.0 allows the currency of the account to be included in the Refresh record. Amount fields in the PBF are assumed to be in the currency specified by the Account currency field. A single PBF Refresh file may contain amounts specified in multiple currencies.

Transactions that impact the PBF are logged to the PTLF with amounts in the Transaction currency, which may differ from the Account currency. When transaction impacting is performed by the Refresh process, the currency in the PTLF is compared with the Account currency in the PBF and conversion is performed if necessary. Therefore, the Refresh process always uses the Account currency from the PBF record to ensure that impacting from the PTLF is correctly valued.

CAF Refresh

For Cardholder Authorization File (CAF) Refreshes, the amounts in the input file are assumed to be in the currency of the institution that issued the card. The value of the Issuer Institution currency does not have to be the same for all institutions belonging to the same Refresh Group. A single CAF Refresh file may contain amounts specified in multiple currencies.

POS Device Handler Processing

The SPDH and APACS 30/40 device handler modules support multiple currency processing at a single POS terminal by providing the following functions:

- Accept transactions from a single terminal in different currencies
- Call the Currency Conversion utilities
- Verify that the Transaction currency is valid
- Create the Multiple Currency token
- Allow amounts in BASE24-pos Terminal Data files (PTD) to be stored in different formats

A primary currency, the Terminal currency, is associated with each POS terminal in a BASE24 network, defined by the value of the Currency Code field in the associated PTD record. A terminal processing a transaction in a currency other than its primary currency must include the ISO currency code of the transaction in the request message.

The POS Device Handler modules call the Currency Conversion utilities to convert any amount field between two specified currencies and, as part of the processing, to provide the derived exchange rate between the two currencies.

When a request message is received from a terminal, the Transaction currency is used when formatting amounts and limits in the POS Standard Internal Message (PSTM). The Device Handler modules call the Currency Conversion utilities to convert the PTD limits and set the currency code in the PSTM to the Transaction currency. This function allows the POS Router/Authorization module to check PTD limits.

In incoming messages, POS Device Handler modules call the Currency Conversion utilities to verify that the Transaction currency code is valid by checking it against currency codes in the ERF. This function allows POS Device Handlers to reject a message containing an invalid currency code before the transaction reaches the POS Router/Authorization module.

When the Transaction currency code is compared to the Terminal currency code, the Device Handler module creates the Multiple Currency token and stores both the Terminal currency code and the Transaction currency code. This function allows the transaction amounts to be stored in the token in the Transaction currency, for subsequent use during transaction processing.

In a single currency system, all amounts stored in a PTD record are represented in the same currency. In a multiple currency system, PTD records can contain totals in the Terminal currency or as hash totals. The value of the MC TOTALLING TYPE field on PTD screen 2 determines which format the totals are in. If amounts are to be stored in the Terminal currency, the Device Handler module calls the Currency Conversion utilities to convert the transaction amount from the Transaction currency to the Terminal currency. If hash totals are to be stored, no conversion is necessary.

ACI Worldwide Inc.

Section 2

Currency Conversion Processing

The BASE24-pos product supports transaction processing in multiple currencies by defining currency types and allowing these to be used and defined in different currencies in data files and financial messages within BASE24. This section describes these currency types and how currency conversion is performed.

Currency Types

The following currency types are defined for the Multiple Currency product:

Transaction Currency — The currency in which the transaction is delivered to BASE24 from a device, a host, or an interchange. If the transaction enters BASE24 through an interchange, this currency may not be the same currency as that used at the actual remote device (the Original currency). Transaction currency is the key currency for multiple currency processing. The transaction amounts are converted from the Transaction currency (using the Base currency) to any of the other currency types. Transaction currency is the currency in which the amounts are logged to the PTLF.

Acquirer Institution Currency — The default currency for the institution that acquires the transaction, specified by the CRNCY-CDE field in the Institution Definition File (IDF) record for the acquiring institution. Limits defined for the retailer in the POS Retailer Definition File (PRDF) are assumed to be in the Acquirer Institution currency. Limits in the Authorization Selection Table (AST) records are defined in the currency of the institution (FIID) to which the routing information pertains, which is sometimes an acquiring institution.

Issuer Institution Currency — The default currency for the institution that is the issuer for the transaction, specified by the CRNCY-CDE field in the IDF record for the issuing institution. Limits defined in the Card Prefix File (CPF), Cardholder Account File (CAF), and Usage Accumulation File (UAF) are all assumed to be in the Issuer Institution currency. All amounts accumulated in the CAF and UAF are also maintained in Issuer Institution currency. Limits in the AST are in the Issuer Institution currency if the selected record is associated with an issuing institution.

Account Currencies — The designated currency for the account to which the transaction applies. If account details are held on BASE24, records in the Positive Balance File (PBF) hold all balances, limits, and other financial amounts in the Account currency, which is designated by the CRNCY-CDE field in the PBF record.

Base Currency — The currency against which all exchange rates are stored in the Exchange Rate File (ERF) and through which all currency conversion takes place using the currency conversion utilities. The Base currency is specified in the BASE-CURRENCY-CODE param in the LCONF and the Base currency code in the COBNAMES file.

Logical Network Currency — Logical Network currency is maintained in the Multiple Currency product primarily to provide compatibility with standard BASE24 processing for a number of subsidiary functions. This currency type is set by the CURRENCY-CODE param in the LCONF.

Terminal Currency — The currency for the POS device at which the transaction is performed. Limits held in the BASE24-pos Terminal Data files (PTD) are assumed to be in the Terminal currency. Totals held in the PTD are either hash totals or in the Terminal currency, depending on the value of the MC TOTALLING TYPE field. This currency type is not explicitly used within BASE24-pos processing.

Original Currency — The currency used at the POS device at which the transaction originates. If the transaction is acquired by BASE24, Original currency is the same as the Transaction currency. However, if the transaction enters BASE24 through an interchange, the Original currency may be different from any currency known to BASE24. Information can be supplied by an Interchange Interface process in the Original Currency 6.0 token. If the token is present, it can be logged to the POS Transaction Log File (PTLF) and forwarded to hosts. This currency type is not explicitly used within BASE24-pos processing.

Currency Types Held in BASE24-pos Files

Within the BASE24 system, values held in the database are used during transaction processing to check limits and to accumulate usage amounts. The table below lists the files and the currency types held in those files.

File	Description	Currency Type
IDF	The CURRENCY CODE field in the IDF specifies the default institution currency, as defined in the COBNAMES File. Depending on processing, this is the default currency of either the acquirer or issuer institution.	Acquirer Institution currency or Issuer Institution currency
CPF	Values for limit fields	Issuer Institution currency
CAF	Values for limit and accumulator fields	Issuer Institution currency
PBF	Values for amount fields	Account currency
AST	Values for floor limits used to define authorization routing	Acquirer Institution currency or Issuer Institution currency
PRDF	Values for ceiling limits	Acquirer Institution currency
PTD	Values for limit and total fields	Terminal currency for limits fields; totals can be in Terminal currency or hash totals
UAF	Values for accumulator fields	Issuer Institution currency

Currency Conversion

Conversion between two currencies takes place using exchange rates held against a user-selected Base currency and follows appropriate conversion rules. The Base currency is a currency that is selected for the system at installation time and that is unlikely to change. For example, in the case of customers using the Multiple Currency product to support the European Single Currency, the Base currency is always the Euro. If compliance with the European Union is not required, any currency may be used as the Base currency and exchange rates are at the discretion of the institution.

Exchange rates for each of the currencies to be used in the system are held in the Exchange Rate File (ERF), with one rate for each currency supported. Exchange rates can be entered and updated by the user. Rates must be stored for every supported currency; that is, for each currency in which transactions can be processed by BASE24-pos or in which accounts are to be held in BASE24-pos.

The exchange rates are held in terms of the rate at which the Base currency is converted to a specific currency unit. Exchange rates are held to six significant figures, and rounding is performed to adhere to the conversion precision used by Europay and Visa.

Currency conversion is performed using the triangulation method, in which conversion between two currencies uses exchange rates held against the Base currency. For example, if the Base currency is the Euro, converting from Italian lire to French francs is achieved by converting the Italian lire to the Euro and converting the Euro to French francs.

Note: The derived approximate conversion rate (e.g., lire to francs) is included in the Multiple Currency token and logged to the PTLF for information only, but it is not used for processing due to the possibility of rounding inaccuracies.

Although transactions received from international interchanges may have originally been performed in a number of different currencies, these rates are converted by the interchange into one of a small number of agreed currencies for delivery to BASE24-pos. Exchange rates only need to be maintained in the ERF for the agreed Transaction currencies. The Original currency may also be forwarded by the interchange and logged for information purposes only.

No historical rates are stored in the ERF. However, a history of previous exchange rates can be derived by analyzing data in the Online Maintenance File (OMF).

Currency Conversion Utilities

To support multiple currency processing, any process (e.g., the Router/Authorization module) requiring currency conversion calls the Currency Conversion utilities to convert financial amounts from one currency to another, using the exchange rates from the ERF. The converted amounts, and associated information, are held in the Multiple Currency token. See the ***BASE24 Tokens Manual*** for the format of the Multiple Currency token.

The Currency Conversion utilities also manipulate multiple currency information in the BASE24-pos Standard Internal Message (PSTM) and the Multiple Currency token. The Multiple Currency token enables the amount fields to be represented in a variety of currencies.

When the Router/Authorization module is initialized, the Currency Conversion utilities use a prebuilt extended memory table (ERFEMT) to hold information from the Exchange Rate File (ERF). When you need to add or update exchange rates in the ERF, perform the following steps:

1. Update the ERF.
2. Run the Extended Memory Table Build process to update the ERF extended memory table.
3. Warmboot processes that use the ERF extended memory table (e.g., the Device Handler/Router/Authorization process) to ensure it uses the updated ERF extended memory table.

Note: The conversion procedures are designed to meet the requirements of European Union Council Regulation (EC) No. 1103/97. To be compliant, the Euro must be used as the Base currency. In addition, the fixed conversion rates between the Euro and currencies of member states adopting the Euro must be as defined by European Union Council Regulation (EC) No. 2866/98.

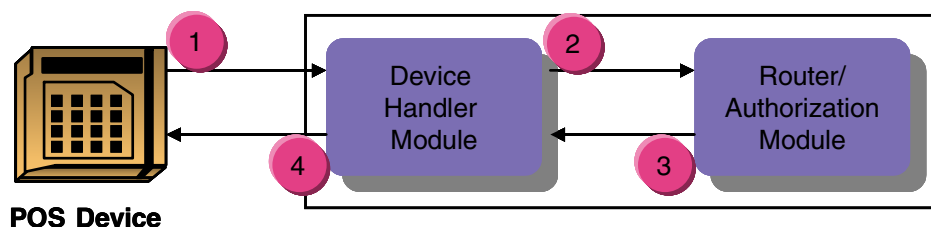
Currency Conversion Flows

This section illustrates the use of the currency types during multiple currency processing.

Currency conversion is performed by the Currency Conversion utilities, which are called by the BASE24 Router/Authorization module during transaction processing.

BASE24-pos Authorizes the Transaction

The following diagram illustrates the currency conversion flow for a multiple currency transaction that originates at a POS device and is authorized by BASE24.

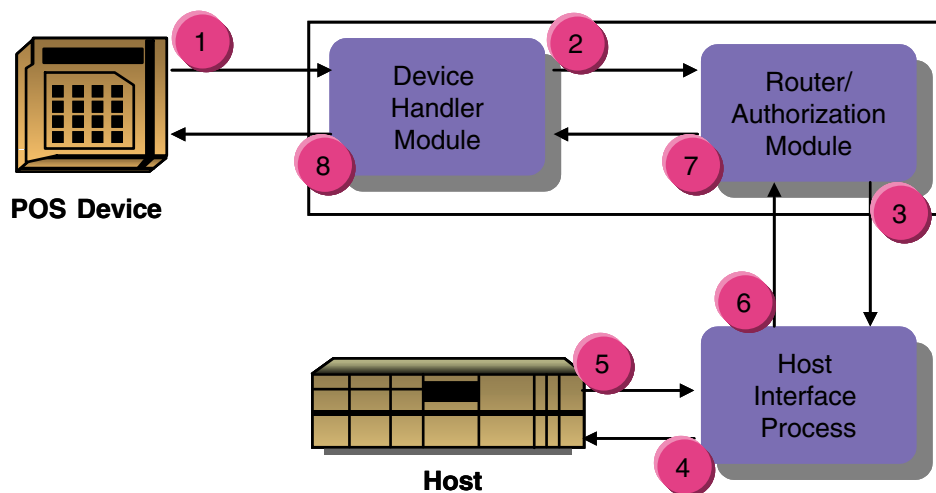


Step	Processing
1	The POS device sends a transaction to BASE24.
2	The Device Handler module looks for a currency code in the native-mode message sent by the device. If a currency code exists, this currency becomes the Transaction currency. If a currency code does not exist in the message, the Device Handler module reads the Terminal currency code from the PTD record. In this case, the Terminal currency from the PTD becomes the Transaction currency. The Device Handler module calls the Currency Conversion utilities to create the Multiple Currency token and sends the transaction to the Router/Authorization module after creating the PSTM.

Step	Processing
3	The IDF contains the currency code for the Issuer Institution currency and the Acquirer Institution currency. Limits, usages, and balances checked in the CPF, IDF, CAF, and UAF are assumed to be in the Issuer Institution currency. If the Transaction currency is not the same as the Issuer Institution currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Issuer Institution currency before limits are checked and before usages and balances are updated. Limits checked in the PRDF are assumed to be in the Acquirer Institution currency. If the Transaction currency is not the same as the Acquirer Institution currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Acquirer Institution currency before limits are checked. Balances, limits, and usages in the PBF are in the Account currency. This currency may or may not be the same as the Issuer Institution currency. If the Transaction currency is not the same as the Account currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Account currency before limits are checked and before usages and balances are updated. PBF balances are returned in the POS Balances token. Balances in both the Account currency and the Transaction currency are moved to the token. If the Transaction currency is not the same as the Account currency, the PBF balances are converted to the Transaction currency. The Router/Authorization module logs the transaction and the Multiple Currency and POS Balances token (if configured to do so) to the PTLF and sends the transaction response to the Device Handler module.
4	The Device Handler module calls the Currency Conversion utilities to convert the transaction amount to the primary Terminal currency if both of the following conditions occur: • The Transaction currency is not the same as the primary Terminal currency. • The MC TOTALLING TYPE field in the PTD indicates that totals are to be stored in the primary Terminal currency. The Device Handler module updates the PTD totals and sends the transaction response to the POS device.

A Host Authorizes the Transaction

The following diagram illustrates the currency conversion flow for a transaction that originates at a POS device and is authorized by a host.



Step

Processing

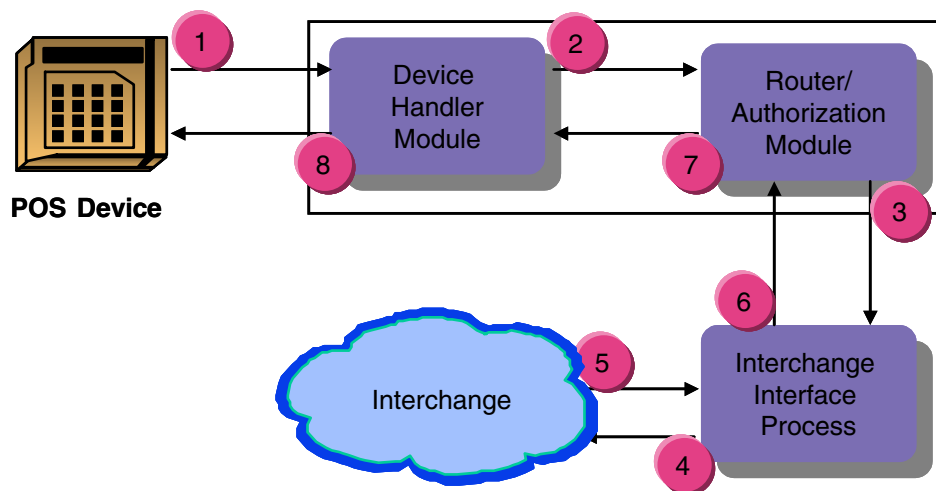
- 1 The POS device sends a transaction to BASE24.
- 2 The Device Handler module looks for a currency code in the native-mode message sent by the device. If a currency code exists, this currency becomes the Transaction currency. If a currency code does not exist in the message, the Device Handler module reads the Terminal currency code from the PTD record. In this case, the Terminal currency becomes the Transaction currency.

The Device Handler module calls the Currency Conversion utilities to create the Multiple Currency token and sends the transaction to the Router/Authorization module.

Step	Processing
3	The Issuer Institution currency and Acquirer Institution currency are specified in the IDF. Limits, usages, and balances checked in the CPF, IDF, CAF, and UAF are assumed to be in the Issuer Institution currency. If the Transaction currency is not the same as the Issuer currency, the Router/Authorization module calls the Currency Conversion utilities to convert the transaction amount from the Transaction currency to the Issuer Institution currency before any limits prescreening checks are performed. Limits checked in the PRDF are assumed to be in the Acquirer Institution currency. If the Transaction currency is not the same as the Acquirer Institution currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Acquirer Institution currency before limits are checked. The Router/Authorization module sends the transaction with the Multiple Currency token to the Host Interface process.
4	The Host Interface process sends the transaction including the Multiple Currency token to the host.
5	The host authorizes the transaction and returns it to the Host Interface process.
6	The Host Interface process sends the transaction to the Router/Authorization module.
7	The Router/Authorization module logs the transaction to the PTLF along with the Multiple Currency token and sends the transaction to the Device Handler module.
8	The Device Handler module calls the Currency Conversion utilities to convert the transaction amount to the primary Terminal currency if both of the following conditions occur: • The Transaction currency is not the same as the primary Terminal currency. • The MC TOTALLING TYPE field in the PTD indicates that totals are to be stored in the primary Terminal currency. The Device Handler module updates the PTD totals and sends the transaction to the POS device.

An Interchange Authorizes the Transaction

The following diagram illustrates the currency conversion flow for a transaction that originates at a POS device and is authorized by an interchange.



Step

Processing

- 1 The POS device sends a transaction to BASE24.
- 2 The Device Handler module looks for a currency code in the native-mode message sent by the device. If a currency code exists, this currency becomes the Transaction currency. If a currency code does not exist in the message, the Device Handler module reads the Terminal currency code from the PTD record. In this case, the Terminal currency becomes the Transaction currency.

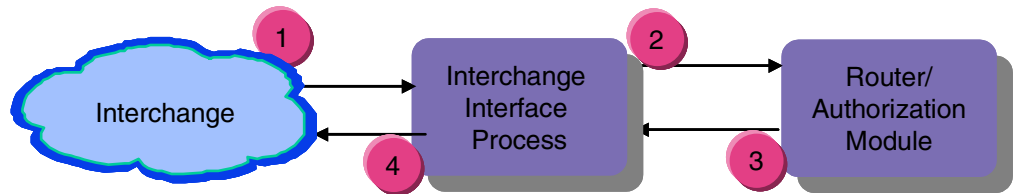
The Device Handler module calls the Currency Conversion utilities to create the Multiple Currency token and sends the transaction to the Router/Authorization module.

Step	Processing
3	The Issuer Institution currency and Acquirer Institution currency are specified in the IDF. (In this case, the issuer is assumed to be the interchange.) Limits, usages, and balances checked in the CPF, IDF, CAF, and UAF are assumed to be in the Issuer Institution currency. If the Transaction currency is not the same as the Issuer Institution currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Issuer Institution currency before any limits prescreening checks are performed. Limits checked in the PRDF are assumed to be in the Acquirer Institution currency. If the Transaction currency is not the same as the Acquirer Institution currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Acquirer Institution currency before limits are checked. The Router/Authorization module sends the transaction with the Multiple Currency token to the Interchange Interface process.
4	The Interchange Interface process suspends the Multiple Currency token and sends the transaction to the interchange.
5	The interchange authorizes the transaction and returns it to the Interchange Interface process.
6	The Interchange Interface process sends the transaction and the Multiple Currency token to the Router/Authorization module.
7	The nonhopper totals in the PTD (e.g., terminal debits, terminal credits) are in the Terminal currency. If the Issuer Institution currency is not the same as the Terminal currency, the Currency Conversion utilities convert the transaction amount to the Terminal currency. The Router/Authorization module logs the transaction to the PTLF along with the Multiple Currency token and sends the transaction to the Device Handler module.

Step	Processing
8	The Device Handler module calls the Currency Conversion utilities to convert the transaction amount to the primary Terminal currency if both of the following conditions occur: • The Transaction currency is not the same as the primary Terminal currency. • The MC TOTALLING TYPE field in the PTD indicates that totals are to be stored in the primary Terminal currency. The Device Handler module updates the PTD totals and sends the transaction to the POS device.

An Interchange Is the Acquirer

The following diagram illustrates the currency conversion flow for a transaction that originates at an interchange and is authorized by BASE24.



Step

Processing

- 1 An interchange sends a transaction to BASE24.

Some interchanges can provide transaction amounts defined in two currencies. The Original currency is the currency used by the POS device at the original point of entry. The interchange converts the Original currency to the Transaction currency before sending the transaction to the Interchange Interface process but may also send information about the Original currency.

- 2 The Interchange Interface process creates the Original Currency 6.0 token and sends the transaction to the Router/Authorization module.

Note: Only the VisaNet and EPS-Net Interchange Interface processes create the Original Currency 6.0 token.

- | Step | Processing |
|-------------|--|
| 3 | <p>The Router/Authorization module creates the Multiple Currency token and, if necessary, the POS Balances token.</p> <p>The IDF contains the currency code for the Issuer Institution currency and the Acquirer Institution currency.</p> <p>Limits, usages, and balances checked in the CPF, IDF, CAF, and UAF are assumed to be in the Issuer Institution currency. If the Transaction currency is not the same as the Issuer Institution currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Issuer Institution currency before limits are checked and before usages and balances are updated.</p> <p>Limits checked in the PRDF are assumed to be in the Acquirer Institution currency. If the Transaction currency is not the same as the Acquirer Institution currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Acquirer Institution currency before limits are checked.</p> <p>Balances, limits, and usages in the PBF are in the Account currency. This currency may or may not be the same as the Issuer Institution currency. If the Transaction currency is not the same as the Account currency, the Currency Conversion utilities convert the transaction amount from the Transaction currency to the Account currency before limits are checked and before usages and balances are updated.</p> <p>PBF balances are returned in the POS Balances token. Balances in both the Account currency and the Transaction currency are moved to the token. If the Transaction currency is not the same as the Account currency, the PBF balances are converted to the Transaction currency.</p> <p>The Router/Authorization module logs the transaction and the tokens (if configured to do so) to the PTLF and sends the transaction to the Interchange Interface process.</p> |
| 4 | <p>The Interchange Interface process forwards the transaction to the interchange.</p> |

Section 3

Authorization Processing

This section describes some of the effects of Multiple Currency processing on the BASE24-pos Router/Authorization process.

Overview

A transaction can enter BASE24-pos in three ways:

Host	The currency of a transaction received from a host is identified by a currency code in the message, and this currency thus becomes the Transaction currency.
Interchange	The currency of a transaction received from an interchange is identified by a currency code in the external message, and this currency thus becomes the Transaction currency. Some interchanges can provide transaction amounts in two currencies. The Original currency is the currency of the transaction conducted at the device, which is converted into the Transaction currency by the interchange.
POS device	The currency of a transaction received from a POS device can be in the currency of the device, as defined in the BASE24-pos Terminal Data files (PTD), or can be selected at the point of sale. The currency sent in the request message to the Device Handler module becomes the Transaction currency (if it is a valid currency).

When the Router/Authorization module receives a transaction, the Currency Conversion utilities create the Multiple Currency token (if necessary) and append it to the internal message. In cases where a transaction is delivered to BASE24 from the VisaNet or EPS-Net interchange, the Original Currency 6.0 token is created by the Interchange Interface process and appended to the PSTM for logging to the PTLF.

Limits and accumulator fields held in the Card Prefix File (CPF), the Cardholder Authorization File (CAF), and the Usage Accumulation File (UAF) are held in the Issuer Institution currency. If the Transaction currency is not the same as the Issuing Institution currency, the Currency Conversion utilities perform the conversions required for limit checking.

Accumulators and limits held in the Positive Balance File (PBF) are held in the Account currency. When the Router/Authorization module checks and updates the balances in the PBF, it either uses the relevant currency from the Multiple Currency token or performs currency conversion so that the transaction amounts and the PBF amounts are in the same currency for the necessary processing.

If the transaction is to be authorized by an external issuer, through an interchange, the transaction (including the Multiple Currency token) is routed to the relevant Interchange Interface process.

If the transaction is to be authorized by a host system, the transaction (including the Multiple Currency token) is routed to the relevant Host Interface process.

The Host Interface or Interchange Interface process suspends the Multiple Currency token and returns it in the response to the Router/Authorization module. Thus, no currency information is lost between request and response.

In conformance with the ISO standard, the Amount field included in response messages from hosts and interchanges must be in the same currency as in the original request. If the host wants to return balances in any other currency, these balances must be conveyed in the POS Balances token.

When BASE24 authorizes a transaction, or a response is received by the Router/Authorization module from an external issuer interface, the transaction is written to a log file. The amount fields in the transaction record are logged in the Transaction currency, and if configured, the Multiple Currency, POS Balances, and Original Currency 6.0 tokens are also logged.

Currency Types

The POS Router/Authorization module manages the processing of transactions originating in a variety of currencies, using authorization criteria (limits and accumulators) that can be specified in different currencies, and against accounts that can be held in different currencies.

To support multiple currencies within a single transaction, the Router/Authorization module identifies the different currency types that may be associated with a transaction. This allows the Router/Authorization module, using the Currency Conversion utilities, to perform currency conversion where necessary and to manipulate data structures (typically the PSTM), where appropriate.

The different currency types defined for the Multiple Currency product are described in section 2. The significance of each type with respect to authorization processing is described below.

Transaction Currency — The currency in which the transaction is delivered to BASE24. This is the key currency for multiple currency processing. Transaction amounts are converted from Transaction currency (using the Base currency) to any of the other currency types. Transaction currency is used for checking ceiling and routing limits and as the main PSTM currency when messages are logged to the PTLF or delivered to Device Handler, Host Interface, or Interchange Interface processes.

Acquirer Institution Currency — The default currency for the institution that acquired the transaction. This currency is specified by the CRNCY-CDE field in the Institution Definition File (IDF) record for the acquiring institution. Limits defined for the retailer in the POS Retailer Definition File (PRDF) are assumed to be in the Acquirer Institution currency. Limits in the Authorization Selection Table (AST) records are defined in the currency of the institution (FIID) to which the routing information pertains, which is sometimes an acquiring institution.

Issuer Institution Currency — The default currency for the institution that is the issuer for the transaction. This currency is specified by the CRNCY-CDE field in the IDF record for the issuing institution. Limits defined in the CPF, CAF, and UAF are all assumed to be in Issuer Institution currency. All amounts accumulated in the CAF and UAF are also maintained in Issuer Institution currency. Limits in the AST are in the Issuer Institution currency if the selected record is associated with an issuing institution. The Issuer Institution currency is used within authorization when checking the above limits, for authorization processing

involving the CAF and UAF, for recording in the Multiple Currency token for delivery to Host Interface processes (if required), and for logging to the token for use by other applications.

Account Currency — The designated currency for the account to which the transaction applies. If account details are held on BASE24, records in the PBF hold all balances, limits and other financial amounts in Account currency, which is designated by the CRNCY-CDE field in the record. Account currency is used within authorization for any authorization processing that involves the PBF only, for logging to the Multiple Currency token for use by other applications, and for inclusion in the POS Balances token, where appropriate.

If the PBF is not used, the Account currency is not known to the Router/Authorization module. A POS Balances token, which includes account information, may be returned in a host response message. This can be logged to the PTLF and passed through to the Device Handler module, but no specific Account currency processing is performed.

Base Currency — The currency against which all exchange rates are stored in the ERF and through which all currency conversion takes place using the Currency Conversion utilities. Base currency is not explicitly used within authorization, but is used by the Currency Conversion utilities and is recorded in the Multiple Currency token when the token is created.

Logical Network Currency — Logical Network currency is maintained in the Multiple Currency product to provide compatibility with standard BASE24 processing. Logical Network currency is not explicitly used within POS authorization, but it is added to the Multiple Currency token by the Currency Conversion utilities when the token is created. The value is available for customer-designed reports.

Initialization and Warmboot Processing

When the Router/Authorization module is initialized, the Currency Conversion utilities use a prebuilt extended memory table (ERFEMT) to hold conversion rate information from the Exchange Rate File (ERF).

You can warmboot the Router/Authorization module by entering the DELIVER PROCESS command using either ALTERATTRIBUTEERFEMT or ALTERATTRERFEMT as the text. Warmbooting the ERF extended memory table causes the Currency Conversion utilities to reread information in the ERF extended memory table. The following is an example of a command to warmboot the ERF extended memory table:

```
DELIVER PROCESS P1A^RTAU, "ALTERATTRERFEMT"
```

For more information on extended memory tables, see the ***BASE24-pos Transaction Processing Manual***.

Preauthorization Processing

The Router/Authorization module includes totals of preauthorization hold amounts when authorizing transactions and in calculation of balance information. This preauthorization hold information is held in different authorization files, dependent on the authorization method in use.

- In the PBF, preauthorization holds are maintained in Account currency.
- In the CAF and UAF, preauthorization holds are maintained in Issuer Institution currency.

BASE24 supports options that allow for preauthorization holds to be maintained in both the CAF and PBF. Any hold amounts from the CAF are first converted from Issuer Institution currency to Account currency before being applied to the PBF.

ACI Worldwide Inc.

Index

A

- Account currency
 - defined, [2-2](#)
 - use of, in authorization processing, [3-5](#)
- Acquirer Institution currency
 - defined, [2-2](#)
 - use of, in authorization processing, [3-4](#)
- Authorization levels supported, [1-4](#)
- Authorization processing
 - impacts of Multiple Currency product, [3-2](#)
 - preauthorization hold amounts, [3-7](#)
 - use of currency types in, [3-4](#)

B

- Base currency
 - defined, [2-3](#)
 - in currency conversion, [2-5](#)
 - use of, in authorization processing, [3-5](#)
- BASE-CURRENCY-CODE param, [1-9](#)

C

- CAF Refresh, impacts of Multiple Currency product, [1-11](#)
- CURR CODE field
 - Multiple Referrals screen, [1-7](#)
 - PTLF, [1-6](#)
- CURRENCY CODE field
 - IDF, [1-6](#)
 - PBF, [1-6](#)
 - PRF, [1-7](#)
 - PTD, [1-7](#)
- Currency conversion
 - described, [2-5](#)
 - exchange rates, [2-5](#)
 - message flows, [2-7](#)
- Currency Conversion utilities
 - impacts on authorization processing, [3-2](#)
 - impacts on Device Handler processing, [1-12](#)
 - use of, [2-6](#)
- Currency types
 - defined, [1-2](#), [2-2](#)
 - in BASE24 files, [2-4](#)
 - use of, in authorization processing, [3-4](#)
 - use of, in Device Handler processing, [1-12](#)
- CURRENCY-CODE param, [1-9](#)

D

- DELIVER PROCESS command, used to warmboot the Router/Authorization module, [3-6](#)
- Device Handler processing, [1-12](#)

E

- ERF assign, [1-9](#)
- ERFEMT assign, [1-9](#)
- ERFEMT, *see* Exchange Rate File (ERF) extended memory table
- ERFEMT-MAX-TBL-SIZE param, [1-9](#)
- ERFEMT-WARMIdentifier LCONF param, [1-9](#)
- Exchange Rate File (ERF)
 - ERF LCONF assign, [1-9](#)
 - ERFEMT LCONF assign, [1-9](#)
- Exchange Rate File (ERF) extended memory table
 - used by Currency Conversion utilities, [2-6](#)
 - used by Router/Authorization module, [3-6](#)
- Exchange rates, used in currency conversion, [2-5](#)

H

- Hash totals, stored in BASE24-pos Terminal Data file (PTD), [1-13](#)

I

- ISO Host Interface process, impacts of Multiple Currency product, [1-10](#)
- Issuer Institution currency
 - defined, [2-2](#)
 - use of, in authorization processing, [3-4](#)

L

- Logical Network currency
 - defined, [2-3](#)
 - use of, in authorization processing, [3-5](#)

M

- MC TOTALLING TYPE field, PTD, [1-7](#)
- Message flows, currency conversion
 - BASE24-pos authorizes transaction, [2-8](#)
 - host authorizes transaction, [2-10](#)

Message flows, currency conversion *continued*
 interchange authorizes transaction, [2-12](#)
 interchange is acquirer, [2-15](#)
Message types supported, [1-4](#)
Multiple Currency product, features of, [1-2](#)
Multiple Currency token
 created by Device Handler modules, [1-12](#)

O

Original currency, defined, [2-3](#)

P

PBF Refresh, impacts of Multiple Currency
 product, [1-11](#)
Preauthorization processing, [3-7](#)
PTLF extracts, impacts of Multiple Currency
 product, [1-10](#)

R

Response codes supported, [1-4](#)

T

Terminal currency
 defined, [2-3](#)
 stored in BASE24-pos Terminal Data file
 (PTD), [1-13](#)
 use of, in Device Handler processing, [1-12](#)
Transaction currency
 defined, [2-2](#)
 in Device Handler processing, [1-12](#)
 use of, in authorization processing, [3-4](#)
Transaction types supported, [1-4](#)

W

Warmbooting the ERF extended memory table, [3-6](#)